VaaniFlow

ENTERPRISE VOICE-TO-ACTION RELIABILITY LAYER

EXECUTIVE MEMO // ONE-PAGE BRIEF
VERIFIED QUALITY: **531 / 531 TESTS PASSING**

The Reliability Layer Between Messy Human Speech and Autonomous Enterprise Action

Converts informal, code-mixed Hinglish voice notes into structured, audit-ready ERP and warehouse transactions with **zero unverified writes**.

01 THE PROBLEM: $100B+ TRAPPED IN VOICE NOTES

63M+ SMES

- **Informal Trade Dominance:** Indian commerce across building materials, steel, agrochemicals, and wholesale distribution operates via WhatsApp voice memos, calls, and noisy yard environments.
- **Manual Entry Backlog:** Traditional ERPs require tedious screen-by-screen form entry, creating hours of fulfillment delay and high dispatch error rates.
- **Why Generic Voice AI Fails:** Naive LLMs treat spoken words as facts, invent historical rates, confuse similar customer accounts, miss self-corrections, and trigger blind dispatches.

02 THE ARCHITECTURAL MOAT

ZERO UNVERIFIED WRITES

- **"Speech is Not State":** Spoken prices and quantities are treated as *untrusted claims* grounded against live business ledgers—never accepted as ground truth.
- **Deterministic Compiler (v2.0):** Mathematically resolves mid-sentence corrections (`"200—actually 220"`) and explicit negations (`"don't dispatch"`) before database contact.
- **Cryptographic Snapshot Binding:** Execution plans lock to live database hash (`4eb7ab8b`). If stock, price, or credit limits drift, the plan safely expires (TOCTOU guard).
- **Post-Write Read-Back Verification:** Never trusts API status 200. Independently queries database rows to verify committed records before issuing receipts.

03 NATIVE MESSAGING INGRESS (NO NEW APPS)

TELEGRAM & WHATSAPP

- **Natural Voice Notes:** Yard owners hold the mic on Telegram or WhatsApp and speak in Hinglish. AudioNormalizer converts Opus to 16kHz PCM WAV for Indic ASR.
- **1-Tap Interactive HUD:** Bot replies with structured order details, 28% GST calculation, 8 check results, and inline `[Approve]` / `[Clarify]` buttons.
- **Principal Authentication:** User identity binds to verified `Role.OPERATOR` ; anonymous users are strictly blocked from superseding approvals.

04 THE THREE-LAYER EXECUTION ENGINE

END-TO-END VERIFIED

1. UNDERSTAND	Gnani Vachana STT + SemanticCompiler v2.0 extracts untrusted claims, slots, revisions, and relative dates (`"parso"` → Day -2).
2. VALIDATE	Grounds CUST-001 & Quote Q-0030 (≤ ₹420/bag). Runs 8 deterministic checks (GST, stock, credit). Locks cryptographic snapshot hash.
3. EXECUTE	Authenticated owner approval triggers idempotent `create_order` → `create_invoice` → `dispatch_order` with read-back proof.

05 MARKET & UNIT ECONOMICS

$12B+ B2B WORKFLOW

- **Target Segments:** High-volume wholesale distributors, stockists, and logistics yards (cement, steel, agrochemicals, FMCG) in hubs like Peenya (BLR), Surat, and Chawri Bazar.
- **Monetization:** Usage-based SaaS (₹3 – ₹8 per verified voice dispatch transaction) + enterprise ERP integration connectors.
- **Defensibility:** Deep operational integration into daily messaging habits paired with tamper-evident audit lineage for accounting.

06 BENCHMARK SCORECARD & RELIABILITY PROOF

PROVABLE SAFETY

10 / 10	0.0%	531	10x
BENCHMARK (A–J)	FALSE EXECUTION	PASSING TESTS	CYCLE SPEEDUP

Tested against 10 canonical failure modes: ASR noise, expired quotes, stock shortages, GST rule conflicts, and mid-sentence customer corrections.

Core Thesis: People speak in messy, contextual language. Enterprise databases require immutable truth. **VaaniFlow bridges the two.**

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